Effect of dark chocolate on renal tissue oxygenation as measured by BOLD-MRI in healthy volunteers.

BACKGROUND: Cocoa is rich in flavonoids, has anti-oxidative properties and increases the bioavailability of nitric oxide (NO). Adequate renal tissue oxygenation is crucial for the maintenance of renal function. The goal of this study was to investigate the effect of cocoa-rich dark chocolate (DC) on renal tissue oxygenation in humans, as compared to flavonoid-poor white chocolate (WC). METHODS: Ten healthy volunteers with preserved kidney function (mean age ± SD 35 ± 12 years, 70% women, BMI 21 ± 3 kg/m2) underwent blood oxygenation level-dependent magnetic resonance imaging (BOLD-MRI) before and 2 hours after the ingestion of 1 g/kg of DC (70% cocoa). Renal tissue oxygenation was determined by the measurement of R2* maps on 4 coronal slices covering both kidneys. The mean R2* (= 1/T2*) values in the medulla and cortex were calculated, a low R2* indicating high tissue oxygenation. Eight participants also underwent BOLD-MRI at least 1 week later, before and 2 hours after the intake of 1 g/kg WC. RESULTS: The mean medullary R2* was lower after DC intake compared to baseline (28.2 ± 1.3 s-1 vs. 29.6 ± 1.3 s-1, p = 0.04), whereas cortical and medullary R2* values did not change after WC intake. The change in medullary R2* correlated with the level of circulating (epi)catechines, metabolites of flavonoids (r = 0.74, p = 0.037), and was independent of plasma renin activity. CONCLUSION: This study suggests for the first time an increase of renal medullary oxygenation after intake of dark chocolate. Whether this is linked to flavonoid-induced changes in renal perfusion or oxygen consumption, and whether cocoa has potentially renoprotective properties, merits further study.